

Day 2

Task 1: Preprocessing Plan and Implementation

Numeric and categorical features were listed explicitly, and remaining question marks were confirmed converted to null values. A ColumnTransformer pipeline was built with two branches. The numeric branch uses median imputation followed by standard scaling. Median was chosen over mean because columns like capital gain and capital loss are heavily skewed with many zeros and a few large outliers, so the mean would be pulled off center while the median stays typical. The categorical branch uses most frequent imputation followed by one hot encoding, since these columns have no natural order and one hot avoids implying a false ranking. Target encoding and dropping rows with missing values were both considered and skipped for now, to keep the pipeline simple and avoid extra leakage risk.

Task 2: Training Two Supervised Models

Two pipelines were built, each combining the same preprocessor with a different estimator: Logistic Regression with a liblinear solver and a fixed random state, and a Decision Tree Classifier with a fixed random state. Both were fit on the training set only, keeping the hold out test set untouched until evaluation.

Task 3: Evaluation on the Hold-out Test Set

Both models were scored on the same hold out test set used on Day 1, alongside the Day 1 rule based baseline for comparison. Accuracy, precision, recall, F1, ROC AUC, and PR AUC were computed for each, along with ROC curves, precision-recall curves, and confusion matrices.

The full comparison table with exact numbers lives in the notebook, alongside the ROC and precision-recall curves. Both supervised models beat the Day 1 rule-based baseline on precision and ROC AUC. Since precision is the primary metric, false positives matter most, and both models keep those relatively low compared to the simple baseline. False negatives are more common than false positives for both models, which is the right trade-off given our goal.

Task 4: Interpretability Check

For Logistic Regression, coefficients were mapped back to their one hot encoded feature names. The top positive coefficients were features like being married, holding an executive or professional occupation, higher education, and higher capital gain, all intuitively linked to higher income. The top negative coefficients were features like being in school or never married and working in service occupations, all intuitively linked to lower income. For the Decision Tree, the top splits keyed on capital gain, education level, and marital status, closely matching the logistic regression's top features. The tree showed a clear gap between train and test accuracy, a sign of overfitting that an unconstrained tree is prone to.

Task 5: Write-up and Model Selection for Day 3

Logistic Regression is the primary model going into Day 3. It matches or beats the Decision Tree on precision and ROC AUC while being much easier to explain, which matters for a business use case where decisions may need to be justified. The Decision Tree is kept as a secondary candidate for its ability to capture non-linear patterns, but only after addressing its overfitting, most likely by limiting tree depth or minimum leaf size. Both models already beat the Day 1 baseline, confirming that proper preprocessing and real models were worth the extra work. The preprocessing pipeline and both fitted model pipelines were saved for reuse tomorrow.

Preprocessing changes to test tomorrow:

- Try target or frequency encoding for high cardinality columns like native country and occupation
- Add a max depth or minimum leaf size constraint to the Decision Tree to reduce overfitting
- Try log-transforming capital gain and capital loss instead of just scaling them
- Test simple feature interactions, such as hours worked combined with occupation